

Q.1. Answer the following short questions.

(6x5=30)

- I. Explain the concept of Calculus of Variations.**
- II. Find Particular Solution of: $y^{(4)}(t) + 6y'''(t) + 14y''(t) + 16y'(t) + 8y = 2$**
- III. Find characteristic roots of: $(r + 3)(r + 3)(r^2 + 2r + 2) = 0$**
- IV. Write complementary function of characteristic equation $(r + 3)(r + 3)(r^2 + 2r + 2) = 0$.**
- V. Explain the concept of Primal and Dual.**
- VI. Explain The Maximum Principal.**

Answer the following questions.

(3x10=30)

- Q. 2. Test convergence of the path of $y_{t+2} - 4y_{t+1} + y_t = 12$, using Schur Theorem.**
- Q. 3. Elaborate Inflation-Unemployment Model using Simultaneous Differential and Difference Equations.**
- Q. 4. Write Kuhn Tucker Conditions for numerical, $U = xy$, $x + y \leq 100$, $x \leq 40$, $x, y \geq 40$. Further solution not required.**

I. Explain the concept of Calculus of Variations

Calculus of Variation

The calculus of variation is used in economics, it is mostly used in economics Mathematical Question.

It can be written as

$$J = \int_0^T F(t, y(t), y'(t)) dt$$

$$\begin{aligned} y(T) &= A \text{ (A given)} \\ y(0) &= g \text{ (A' given)} \end{aligned}$$

It is the fundamental problem of calculus of variation.

It is necessary that it must be marginal convergent.

In calculus of variation equations are

In calculus of variation Euler Equation also used.

Euler equation:-

- The first basic condition of calculus of variation is an Euler Equation. Euler Equation is mostly used in calculus of variation.

Euler Equation play important Role in calculus of variation Question.

It is can be written as.

$$\therefore \frac{\partial F}{\partial x} = \frac{d}{dt} \left(\frac{\partial F}{\partial \dot{x}} \right)$$

We can Explain Euler Equation through diagram.

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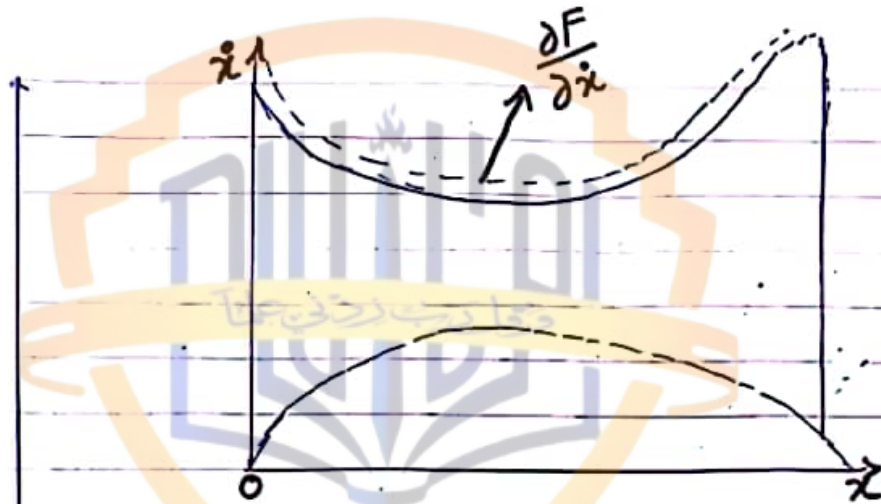
$$\begin{aligned} y(T) &= A \text{ (A given)} \\ y(0) &= g \text{ (A' given)} \end{aligned}$$

It is the fundamental problem of calculus of variation.

It is necessary that it must be marginal convergent.

In calculus of variation equations are continuous differential equations.

Maximization and Minimization is (same) difference requirement but share the same characteristic roots.



Steps in (Euler) calculus of
Variation:-

Step 1:- $F = t \cdot y(t) ; y'(t)$

Step:- Taking derivative w.r.t
 $\frac{\partial F}{\partial x}$ and $\frac{\partial F}{\partial \dot{x}}$

ep 3:-

put both derivative in Euler equation.

$$\therefore \frac{\partial F}{\partial x} = \frac{d}{dt} \left(\frac{\partial F}{\partial \dot{x}} \right)$$

Step 4:-

If $\ddot{x}$ is required taking integration of the derivative.

Step 5:-

put limit in the equation

Step 6:-

Then (taking) find sufficient condition

$$D^2 = \begin{vmatrix} F_{xx} & F_{x\dot{x}} \\ F_{x\dot{x}} & F_{\dot{x}\dot{x}} \end{vmatrix} \quad D^2 = \begin{vmatrix} F_{\dot{x}\dot{x}} & F_{x\dot{x}} \\ F_{x\dot{x}} & F_{xx} \end{vmatrix}$$

II. Find Particular Solution of: $y^{(4)}(t) + 6y'''(t) + 14y''(t) + 16y'(t) + 8y = 2$

Answer:

$$y^{(4)}(t) + \underbrace{6}_{a_1} y'''(t) + \underbrace{14}_{a_2} y''(t) + \underbrace{16}_{a_3} y'(t) + \underbrace{8}_{a_4} y = \underbrace{2}_b$$

Here; $a_1 = 6, a_2 = 14, a_{n-1} = a_3 = 16, \underbrace{a_n = a_4 = 8 \neq 0}_{y_p = \frac{b}{a_n}}$ and $\underbrace{b = 2 \neq 0}_{\text{non-homogenous case}}$.

$$y_p = \frac{2}{8}$$

$$y_p = \frac{1}{4}$$

$$\boxed{y_p = \frac{1}{4}}$$

Interpretation: The 4th (higher) order differential equation,

$$y^{(4)}(t) + \underbrace{6}_{a_1} y'''(t) + \underbrace{14}_{a_2} y''(t) + \underbrace{16}_{a_3} y'(t) + \underbrace{8}_{a_4} y = \underbrace{2}_b$$

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$$y_p = \frac{2}{8}$$

$$y_p = \frac{1}{4}$$

$$\boxed{y_p = \frac{1}{4}}$$

Interpretation: The 4th (higher) order differential equation, $\boxed{y^{(4)}(t) + 6y'''(t) + 14y''(t) + 16y'(t) + 8y = 2}$ has a particular integral $\boxed{y_p = \frac{1}{4}}$. It is a **static equilibrium** as y_p is a constant value.

III. Find characteristic roots of: $(r + 3)(r + 3)(r^2 + 2r + 2) = 0$

Answer:

$$\underbrace{(r + 3)}_{\text{linear function}} \times \underbrace{(r + 3)}_{\text{linear function}} \times \underbrace{(r^2 + 2r + 2)}_{\text{quadratic equation}} = 0$$

Either;

$(r + 3) = 0$	$(r + 3) = 0$	$\left(\overset{a}{\widetilde{(1)}} r^2 + \overset{b}{\widetilde{2}} r + \overset{c}{\widetilde{2}} \right) = 0$
$\Downarrow$	$\Downarrow$	$r_3, r_4 = \frac{-b \pm \sqrt{b^2 - 4ac}}{2}$
$\Downarrow$	$\Downarrow$	$r_3, r_4 = \frac{-2 \pm \sqrt{2^2 - 4(1)(2)}}{2}$
$r_1 = -3$	$r_2 = -3$	$r_3, r_4 = \underbrace{-1}_h \pm \underbrace{1}_v i$

Summarizing the (hybrid) group of roots:

Summarizing the (hybrid) group of roots.

$$\underbrace{r_1 = -3, r_2 = -3}_{\text{Repeated real roots}}, \underbrace{r_3 = -1 + i, r_4 = -1 - i}_{\text{conjugate complex roots}}$$

Interpretation: Complementary Function in case of higher order differential equations is hybrid. Here, it is a combination of Repeated Real roots and Complex roots.

IV. Write complementary function of characteristic equation $(r + 3)(r + 3)(r^2 + 2r + 2) = 0$.

Answer: Since the roots are already found in last part:

Hybrid group of roots:

$$\underbrace{r_1 = -3, r_2 = -3}_{\text{Repeated real roots}}, \underbrace{r_3 = -1 + i, r_4 = -1 - i}_{\text{conjugate complex roots}}$$

Accordingly, the complementary function will also be hybrid:

<i>Hybrid nature of complementary function</i>	
$y_c = \underbrace{A_1 e^{r_1 t} + A_2 e^{r_2 t}}_{\text{Repeated Real roots segment}}$	$+ \underbrace{e^{ht} (A_3 \cos vt + A_4 \sin vt)}_{\text{Complex roots segment}}$

$$2) = 0.$$

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Hybrid group of roots:

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$$y_c = A_1 e^{-3t} + A_2 e^{-3t} + e^{(-1)t} \{A_3 \cos(1)t + A_4 \sin(1)t\}$$

Hybrid nature of complementary function

$$y_c = \underbrace{A_1 e^{-3t} + A_2 t e^{-3t}}_{\text{Repeated Real Roots}} + \underbrace{e^{-t}(A_3 \cos t + A_4 \sin t)}_{\text{Complex Roots}}$$

V. Explain the concept of Primal and Dual.

Explain concept of primal & dual?

The original problem in linear Programming Problem is known as primal, whereas, its counter-part is known as dual.

Every primal in LPP has associated with it an another program known as dual.

Corresponding their minimization, there always exists a counter part maximization such as:
$$\bar{C} = \bar{C}^*$$

And every maximization problem, there is always exists a corresponding counter part minimization

as $\pi = \pi^*$

Example:

Primal	Dual
Max: $\pi = C_1 x_1 + C_2 x_2$	Min: $\pi^* = r_1 y_1 + r_2 y_2$
Subject $\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \leq \begin{bmatrix} r_1 \\ r_2 \end{bmatrix}$	Subject $\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \geq \begin{bmatrix} C_1 \\ C_2 \end{bmatrix}$
To: $\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \leq \begin{bmatrix} r_1 \\ r_2 \end{bmatrix}$	To: $\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \geq \begin{bmatrix} C_1 \\ C_2 \end{bmatrix}$
Non negativity: $x_1, x_2 \geq 0$	Non negativity: $y_1, y_2 > 0$

The optimal values of primal & dual constraints always exists

Duality Th:

1st Th:- The optimal value of primal & duals always converge & do exist.

$$\bar{C} = \bar{C}^* \quad \& \quad \bar{A} = \bar{A}^*$$

2nd Th: If certain choice variables are non-zero, then their corresponding dummy variables are zero.

such as, $y_i > 0 \Rightarrow S_i = 0$

$$x_i > 0 \Rightarrow t = 0$$

And if dummy variables are ^{non-}zero, then their corresponding choice variable are zero

As: $S_i > 0 \Rightarrow y_i = 0$

$$t > 0 \Rightarrow x_i = 0$$

VI Explain The Maximum Principle

VI. Explain The Maximum Principal.

Maximum principal

Optimal control Theory :-

Optimal control Theory is the problem taking in dynamic optimization.

Optimal control Theory (control) focus upon one or more variable.

Optimal control theory find in state

y and control Theory U .

In optimal control Theory ^{variable} found $U^* = f(x)$

But we find $\vec{U} = f(x)$

Maximum principle:-

Maximum principle used in optimal control Theory

In optimal control Theory the first order differential equation use in Maximum principle. Statement of maximum principal involve two concept.

(i) Constant variable

(ii) Humiltionion function.

Constant variable:-

These concept already present in the problem.

We turn out to be the study in which the new variable will emerged. it is known as constant variable. It is denoted by "d".

(iii) Humiltionion function:-

(ii) Humiltionian function:-

Another concept use in the Maximum principal is the Humiltionian function. "The entry of the constant variable in optimal control theory it is known as Humiltionian function. It is denoted by H .
It is written as.

$$H = f(t, u, y, d)$$

- In linear programming the single order differential equation state y .
In maximum principle the second order differential equation have two variable state variable " y " and the control

variable U .

Steps in Maximum principal:-

Step 1:- $H = f(u, t, y, d)$

Step 2:- derivative w.r.t y, x and d
The derivative of x is the value of d and the derivative of the d is the value of x .

Step 3:-

$$-\frac{\partial H}{\partial y} = -\lambda \text{ (notation of } y)$$

$$\frac{\partial H}{\partial x} = \text{(notation of } \lambda)$$

Step 4:-

Interpretation of the λH - λ

Step 4:-

Integration of the $\frac{\partial H}{\partial x} = -\dot{V}$

Then :-

$$U^* = f(u, t, y, d) \Rightarrow U = f(u, t, y, d)$$

$\dot{U}$ = optimal control variable

U = another control variable.

The Maximum requirement of w.r.t to
y the name arises are the
Maximum principal.

Part II (Long Essay Questions)

Give answers of the following:

(3x10=30)

Q. 2. Test convergence of the path of $y_{t+2} - 4y_{t+1} + y_t = 12$, using Schur Theorem. (10)

Answer: Here: $\underbrace{1}_{a_0} y_{t+2} + \underbrace{(-4)}_{a_1} y_{t+1} + \underbrace{(1)}_{a_2} y_t = \underbrace{12}_c$

$$\Delta_1 = \begin{vmatrix} a_0 & a_2 \\ a_2 & a_0 \end{vmatrix} \leq 0$$

$$\Delta_2 = \begin{vmatrix} a_0 & 0 & a_2 & a_1 \\ a_1 & a_0 & 0 & a_2 \\ a_2 & 0 & a_0 & a_1 \\ a_1 & a_2 & 0 & a_0 \end{vmatrix} \leq 0$$

Here; $a_0 = 1$, $a_1 = -4$, $a_2 = 1$, and $\underbrace{c = 12 \neq 0}_{\text{non-homogenous case}}$.

Recalling two Schur Theorem determinants for second-order difference equation:

$$\Delta_2 = \begin{vmatrix} a_1 & a_0 & 0 & a_2 \\ a_2 & 0 & a_0 & a_1 \\ a_1 & a_2 & 0 & a_0 \end{vmatrix} \leq 0$$

Here; $a_0 = 1$, $a_1 = -4$, $a_2 = 1$, and $c = 12 \neq 0$.
non-homogenous case

Recalling two Schur Theorem determinants for second-order difference equation:

$$\Delta_1 = \begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix} \leq 0$$

$$\Delta_1 = \begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix} = 0$$

$$\boxed{\Delta_1 = 0}$$

First Determinant (Δ_1) is found to be non-negative.

We find the second determinant to see if it violates the requirement Convergence in Schur Theorem which requires all of the determinants to be positive.

Evaluating the second determinant (Δ_2).

$$\Delta_2 = \begin{vmatrix} 1 & 0 & 1 & -4 \\ -4 & 1 & 0 & 1 \\ 1 & 0 & 1 & -4 \\ -4 & 1 & 0 & 1 \end{vmatrix} = 0$$

$$\boxed{\Delta_2 = 0}$$

Second determinant (Δ_2) is also non-negative. This refuses the condition of Convergence. Neither it satisfies the condition of Divergence. Hence, the time path can be considered to have **Uniform Oscillation**.

Inflation Model:-

i) Expectation Argument:-

$$P = X - T - BU + g\pi$$

(ii) Adaptive Expectative:-

$$\frac{\partial F}{\partial t} = F(0-t)$$

(iii) Monetary policy:-

$$\frac{\partial y}{\partial t} = -k(\pi - P)$$

(iv) Endogenous variable:-

P = Actual rate of inflation

π = Present rate of inflation

U = rate of unemployment.

We add the period in differential equation these equation can be following.

Expective Argument

$$P_t = k - T U_{-BUT} + g T t$$

expectation

Adoptive Expectation:-

the derivative replace by the differential equation.

Example:-

$$\pi_{t+1} - \pi = J(P_t - T_t)$$

In monetary policy, derivation is replaced by differential equation.

$$U_{t+1} - U = -k(\pi - P_{t+1})$$

These three expectation contain a new dynamic variable by inflation and unemployment model.

no more parameter are use the program

the derivative replace by the differential equation.

Example:-

$$\pi_{t+1} - \pi = J(P_t - T_t)$$

In monetary policy, derivation is replaced by differential equation.

$$U_{t+1} - U = -k(\pi - P_{t+1})$$

These three expectation contain a new dynamic variable by inflation and unemployment model.

as many parameter are use the programme

“rate of growth of nominal money”

Q. 4. Write Kuhn Tucker Conditions for Numerical, $U = xy, x + y \leq 100, (x, y) \geq 40$. Further solution is not required.

Answer:

Objective Function	<i>product term-nonlinear</i> $\overline{U = xy}$
General Constraints	$x + y \leq 100$
	$x \leq 40$
Nonnegativity Constraints	$x, y \geq 0$

STEP – I: Lagrangian Function:

Introduce the two constraints and hence two multipliers.

$$Z(x, y, \lambda_1, \lambda_2) = \underbrace{xy}_{\text{objective function}} + \underbrace{\tilde{\lambda}_1}_{\text{1st multiplier}} \underbrace{(100 - x - y)}_{\text{1st constraint}} + \underbrace{\tilde{\lambda}_2}_{\text{2nd multiplier}} \underbrace{(40 - x)}_{\text{2nd constraint}}$$

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STEP – II: Kuhn-Tucker Conditions – Maximization:

For two variables (x, y) there are two conditions and for two multipliers (λ_1, λ_2) there are also two conditions.

Variable-Based Conditions			
$\left(\frac{\partial Z}{\partial x}\right) \leq 0$	$x \geq 0$	$\Rightarrow$ either or both (=0)	$x \left(\frac{\partial Z}{\partial x}\right) = 0$
$\left(\frac{\partial Z}{\partial v}\right) \leq 0$	$y \geq 0$	$\Rightarrow$ either or both (=0)	$y \left(\frac{\partial Z}{\partial v}\right) = 0$

STEP – II: Kuhn-Tucker Conditions – Maximization:

For two variables (x, y) there are two conditions and for two multipliers (λ_1, λ_2) there are also two conditions.

Variable-Based Conditions			
$\left(\frac{\partial Z}{\partial x}\right) \leq 0$	$x \geq 0$	$\Rightarrow$ either or both (=0)	$x \left(\frac{\partial Z}{\partial x}\right) = 0$
$\left(\frac{\partial Z}{\partial y}\right) \leq 0$	$y \geq 0$	$\Rightarrow$ either or both (=0)	$y \left(\frac{\partial Z}{\partial y}\right) = 0$
Multiplier-Based Conditions			
$\left(\frac{\partial Z}{\partial \lambda_1}\right) \geq 0$	$\lambda_1 \geq 0$	$\Rightarrow$ either or both (=0)	$\lambda_1 \left(\frac{\partial Z}{\partial \lambda_1}\right) = 0$
$\left(\frac{\partial Z}{\partial \lambda_2}\right) \geq 0$	$\lambda_2 \geq 0$	$\Rightarrow$ either or both (=0)	$\lambda_2 \left(\frac{\partial Z}{\partial \lambda_2}\right) = 0$

$\backslash \sigma \lambda_1 /$	$-$	$both (=0)$	$\backslash \sigma \lambda_1 /$
$\left(\frac{\partial Z}{\partial \lambda_2}\right) \geq 0$	$\lambda_2 \geq 0$	$\Rightarrow$ either or $both (=0)$	$\lambda_2 \left(\frac{\partial Z}{\partial \lambda_2}\right) = 0$

□

Developing First Order Conditions:

$$\frac{\partial Z}{\partial x} = \frac{\delta}{\delta x} \{xy + \lambda_1(100 - x - y) + \lambda_2(40 - x)\} \leq 0$$

$$\frac{\partial Z}{\partial x} = y - \lambda_1 - \lambda_2 \leq 0$$

$$\frac{\partial Z}{\partial y} = \frac{\delta}{\delta y} \{xy + \lambda_1(100 - x - y) + \lambda_2(40 - x)\} \leq 0$$

$$\frac{\partial Z}{\partial y} = x - \lambda_1 \leq 0$$

$$\frac{\partial Z}{\partial \lambda_1} = \frac{\delta}{\delta \lambda_1} \{xy + \lambda_1(100 - x - y) + \lambda_2(40 - x)\} \geq 0$$

$$\frac{\partial Z}{\partial \lambda_1} = 100 - x - y \geq 0$$

$$\frac{\partial Z}{\partial \lambda_2} = \frac{\delta}{\delta \lambda_2} \{xy + \lambda_1(100 - x - y) + \lambda_2(40 - x)\} \geq 0$$

$$\frac{\partial Z}{\partial \lambda_2} = 40 - x \geq 0$$

Converting the inequalities into equalities:

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$$\frac{\partial Z}{\partial \lambda_2} = \frac{\delta}{\delta \lambda_2} \{xy + \lambda_1(100 - x - y) + \lambda_2(40 - x)\} \geq 0$$

$$\frac{\partial Z}{\partial \lambda_2} = 40 - x \geq 0$$

Converting the inequalities into equalities:

$$\frac{\partial Z}{\partial x} = y - \lambda_1 - \lambda_2 \equiv 0$$

$$\frac{\partial Z}{\partial y} = x - \lambda_1 \equiv 0$$

$$\frac{\partial Z}{\partial \lambda_1} = 100 - x - y \equiv 0$$

$$\frac{\partial Z}{\partial \lambda_2} = 40 - x \equiv 0$$

STEP – III: Trial Solution Cases:

$$\frac{\partial \lambda_2}{\partial \lambda_2} = 40 - x \equiv 0$$

STEP – III: Trial Solution Cases:

We try two trial solutions:

Trial Solution Case – I: $\lambda_1, \lambda_2 > 0$	Trial Solution, Case – II: $x, y > 0$
$(\lambda_1 > 0) \wedge \left\{ \lambda_1 \left(\frac{\delta Z}{\delta \lambda_1} \right) = 0 \right. \\ \left. \begin{array}{c} \text{then} \\ \left(\frac{\delta Z}{\delta \lambda_1} \right) = 0 \end{array} \right\}$	$(x > 0) \wedge \left\{ x \left(\frac{\delta Z}{\delta x} \right) = 0 \right. \\ \left. \begin{array}{c} \text{then} \\ \left(\frac{\delta Z}{\delta x} \right) = 0 \end{array} \right\}$

$(\lambda_2 > 0) \wedge \left\{ \lambda_2 \underbrace{\left(\frac{\delta Z}{\delta \lambda_2} \right)}_{\text{then } \left(\frac{\delta Z}{\delta \lambda_2} \right) = 0} = 0 \right\}$	$(y > 0) \wedge \left\{ y \underbrace{\left(\frac{\delta Z}{\delta y} \right)}_{\text{then } \left(\frac{\delta Z}{\delta y} \right) = 0} = 0 \right\}$
$\boxed{\lambda_1, \lambda_2 > 0} \Rightarrow \left\{ \left(\frac{\delta Z}{\delta \lambda_1} \right), \left(\frac{\delta Z}{\delta \lambda_2} \right) \right\} = 0$	$\boxed{x, y > 0} \Rightarrow \left\{ \left(\frac{\delta Z}{\delta x} \right), \left(\frac{\delta Z}{\delta y} \right) \right\} = 0$

Trial Solution Case – I: $\boxed{\lambda_1, \lambda_2 > 0} \Rightarrow \left\{ \left(\frac{\delta Z}{\delta \lambda_1} \right), \left(\frac{\delta Z}{\delta \lambda_2} \right) \right\} = 0$

$\left\{ \left(\frac{\delta Z}{\delta \lambda_1} \right), \left(\frac{\delta Z}{\delta \lambda_2} \right) \right\} = 0$	
$\boxed{\left(\frac{\delta Z}{\delta \lambda_1} \right)} = 0$	$\boxed{\left(\frac{\delta Z}{\delta \lambda_1} \right)} = 0$
$\boxed{(100 - x - y)} = 0$	$\boxed{(40 - x)} = 0$

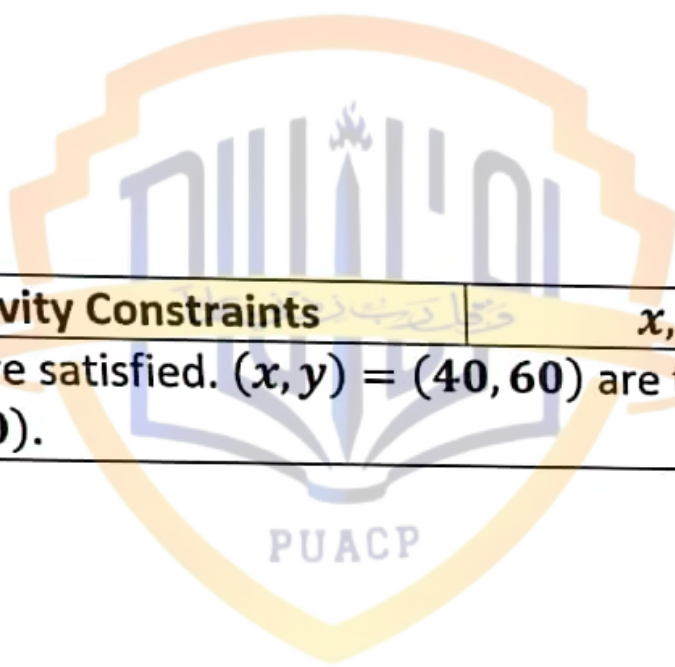
$ (\delta\lambda_1)', (\delta\lambda_2)'\rangle - 0$	
$\left(\frac{\delta Z}{\delta\lambda_1}\right) = 0$	$\left(\frac{\delta Z}{\delta\lambda_1}\right) = 0$
$\underbrace{(100 - x - y)}_{(y=100-x)} = 0$	$\underbrace{(40 - x)}_{x=40} = 0$
$y = 100 - \boxed{x}$ $y = 100 - \boxed{40}$ $y = \boxed{60}$	
Trial Solution Case – I gives us values of both of the variables $(x^*, y^*) = (40, 60)$. Multipliers can also be found.	

For values of λ_1, λ_2 substitute x^* and y^* in $\left(\frac{\delta Z}{\delta x}\right)$ and $\left(\frac{\delta Z}{\delta y}\right)$.	
$\underbrace{\left(\frac{\delta Z}{\delta y}\right)}_{as\ y>0, \left(\frac{\delta Z}{\delta y}\right)>0} = 0$	$\underbrace{\left(\frac{\delta Z}{\delta x}\right)}_{as\ x>0, \left(\frac{\delta Z}{\delta x}\right)>0} = 0$
$\underbrace{x - \lambda_1}_{(x=40)} = 0 \Rightarrow \lambda_1 = 40$	$\underbrace{y - \lambda_1 - \lambda_2}_{\substack{60 \\ 40}} = 0 \Rightarrow \lambda_2 = 20$

$\underbrace{(\overline{\delta y}) - 0}_{as\ y > 0, (\frac{\delta Z}{\delta y}) > 0}$	$\underbrace{(\overline{\delta x}) - 0}_{as\ x > 0, (\frac{\delta Z}{\delta x}) > 0}$
$\underbrace{x - \lambda_1}_{\substack{(x=40) \\ (\frac{\delta Z}{\delta \lambda_2}=0)}} = 0 \Rightarrow \lambda_1 = 40$	$\underbrace{y}_{60} - \underbrace{\lambda_1}_{40} - \lambda_2 = 0 \Rightarrow \lambda_2 = 20$
$(\lambda_1, \lambda_2) = (40, 20)$	

Verification: We found $(x, y) = (40, 60)$ and $(\lambda_1, \lambda_2) = (40, 20)$. Now we check if the values satisfy all of the functions:

Objective Function	$U = xy$	$U = (40)(60) = 2400$
General Constraints	$x + y \leq 100$	$40 + 60 \leq 100$
	$x \leq 40$	$40 \leq 40$

The logo of the Punjab University of Agriculture, Chaudhary Sahib Singh Campus, Ludhiana (PUACP) is visible in the background. It features a shield-like emblem with a book and a torch, surrounded by text in Urdu and English.

Nonnegativity Constraints

$$x, y \geq 0$$

$$40, 60 \geq 0$$

All constraints are satisfied. $(x, y) = (40, 60)$ are the undisputed values at equilibrium that maximize the utility (2400).